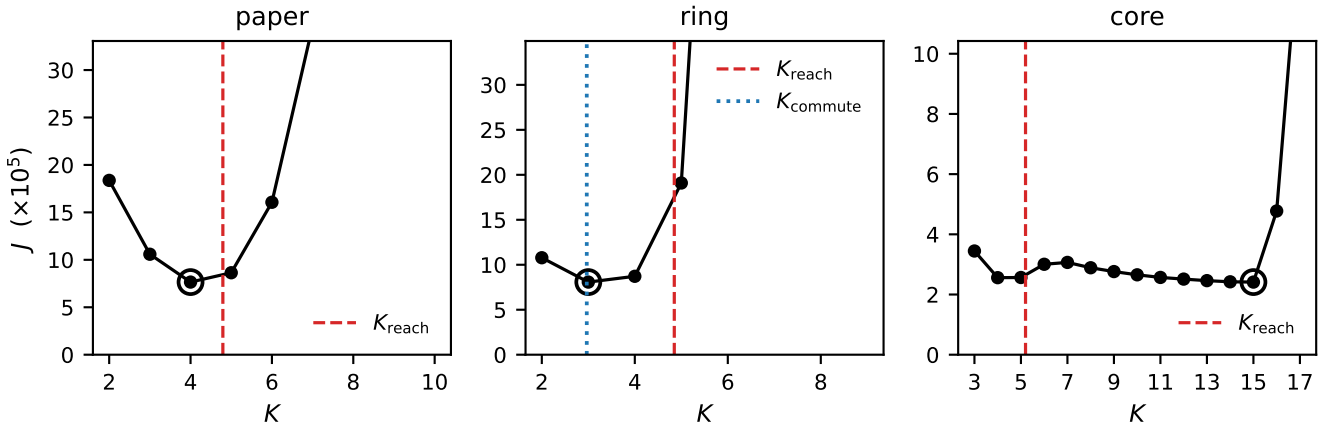
paper
ring
core
J ($\times 10^5$)
K
K_{reach}
K_{commute}